Table 1: E. coli + P. putida datasets (pre-build, 2026-10-14; E. coli + P. putida) built as LMDBs in the TorchCell dev tree. *Genotypes* and *Env* count distinct stored genotype and environment payloads; *Instances* is the record count; *Signal (gzip, bits)* the gzip size in bits of the concatenated stored instances (perturbation + environment + phenotype), as in the yeast table.

Dataset	Host	Genotypes	Env	Instances	Phenotype	Shape	Graph role	Signal (gzip, bits)
Environmental / chemogenomic response								
CrisprPineneToleranceNiu2019	E. coli	54	1	54	environment_response (log2_ratio)	scalar	global	4.0×10^{4}
CrispriChemgenChoe2025	E. coli	39,291	12	470,502	environment_response (log2_ratio)	scalar	global	2.4×10^{8}
CrispriGeneFitnessWang2018	E. coli	13,641	5	20,158	environment_response (log2_ratio)	scalar	global	1.5×10^{7}
CrispriGuideFfaEnrichmentFang2025	E. coli	15,711	1	15,711	environment_response (log2_ratio)	scalar	global	7.3×10^{6}
CrispriGuideFitnessWang2018	E. coli	53,203	5	240,481	environment_response (log2_ratio)	scalar	global	1.6×10^{8}
CrispriKnockdownCui2018	E. coli	71,696	1	143,392	environment_response (log2_ratio)	scalar	global	7.1×10^{7}
CrispriScreenRousset2018	E. coli	40,592	4	92,252	environment_response (log2_ratio)	scalar	global	4.1×10^{7}
DoublingTimeCaglar2017	E. coli	1	16	55	environment_response (growth_rate)	scalar	global	3.2×10^{4}
EnvChemgenGirgis2009	E. coli	3,976	17	66,287	environment_response (z_score)	scalar	global	3.6×10^{7}
EnvChemgenNichols2011	E. coli	3,967	324	1,286,962	environment_response (z_score)	scalar	global	4.0×10^{8}
EnvChemgenShiver2016	E. coli	3,846	51	210,998	environment_response (z_score)	scalar	global	9.9×10^{7}
EnvChemgenWang2015	E. coli	46	1	46	environment_response (log2_ratio)	scalar	global	2.9×10^{4}
EnvChemgenWang2024	E. coli	4,419	6	26,514	environment_response (log2_ratio)	scalar	global	9.3×10^{6}
FitnessBrowserEcoli	E. coli	3,789	3	22,734	environment_response (log2_ratio)	scalar	global	1.1×10^{7}
GrowthRateLamoureux2023	E. coli	21	54	354	environment_response (growth_rate)	scalar	global	1.9×10^{5}
GrowthRateS23Schmidt2016	E. coli	1	22	26	environment_response (growth_rate)	scalar	global	2.0×10^{4}
GrowthYieldBaba2006	E. coli	3,912	3	11,735	environment_response (optical_density)	scalar	global	3.3×10^{6}
MismatchCrispriFitnessHawkins2020	E. coli	24,186	1	24,186	environment_response (relative_growth_rate)	scalar	global	1.5×10^{7}
PhageRbTnseqMutalik2020	E. coli	3,716	71	287,815	environment_response (log2_ratio)	scalar	global	2.1×10^{8}
RbTnseqPrice2018Ecoli	E. coli	3,789	104	598,662	environment_response (log2_ratio)	scalar	global	4.0×10^{8}
RbTnseqTeteneva2024	E. coli	3,691	18	66,162	environment_response (log2_ratio)	scalar	global	2.3×10^{7}
RbTnseqWetmore2015	E. coli	3,646	85	353,662	environment_response (log2_ratio)	scalar	global	2.4×10^{8}
EnvMetalTnseqRoyet2025	P. putida	5,491	4	21,583	environment_response (log2_ratio)	scalar	global	6.7×10^{6}
FitnessBrowserPutida	P. putida	9,556	146	1,118,052	environment_response (log2_ratio)	scalar	global	5.3×10^{8}
IsoprenolSelectionMenasalvas2025	P. putida	58	1	58	environment_response (categorical)	scalar	global	7.5×10^{4}
IsoprenolToleranceLim2025	P. putida	3	2	3	environment_response (log2_ratio)	scalar	global	1.8×10^{4}
LactamGrowthRateThompson2019	P. putida	3	3	9	environment_response (growth_rate)	scalar	global	1.2×10^{4}
RbTnseqBorchert2023	P. putida	271	32	10,824	environment_response (log2_ratio)	scalar	global	4.5×10^{6}
RbTnseqBorchert2024	P. putida	9,464	155	1,466,920	environment_response (log2_ratio)	scalar	global	4.2×10^{8}
Expression (RNA-seq)								
MrnaFractionBalakrishnan2022	E. coli	3	14	28	mrna_number_fraction (rnaseq_mrna_number_fraction)	vector (4342)	node	7.2×10^{6}
RnaseqCaglar2017	E. coli	1	57	152	expression_tpm (rnaseq_tpm)	vector (4196)	node	6.7×10^{7}
RnaseqGuzman2018	E. coli	7	2	14	expression_tpm (rnaseq_tpm)	vector (4329)	node	4.5×10^{6}
RnaseqLamoureux2023	E. coli	74	119	1,035	expression_tpm (rnaseq_tpm)	vector (4257)	node	5.8×10^{8}
RnaseqPublicK12Lamoureux2023	E. coli	72	198	1,675	expression_tpm (rnaseq_tpm_from_released_counts)	vector (4355)	node	9.5×10^{8}
PutidaPrecise321Lim2022	P. putida	14	118	321	expression_tpm (rnaseq_tpm)	vector (5564)	node	2.4×10^{8}
Fitness								
CarbonSourceTong2020	E. coli	3,714	30	111,420	fitness	scalar	global	4.8×10^{7}
CrispriCrossRachwalski2024	E. coli	14,209	15	48,900	fitness	scalar	global	1.9×10^{7}
GrowthAucRapp2026	E. coli	1,515	1	1,515	fitness	scalar	global	7.7×10^{5}
GrowthRateCampos2018	E. coli	3,664	1	3,664	fitness	scalar	global	1.1×10^{6}
GrowthRateChoe2019	E. coli	6	1	8	fitness	scalar	global	1.1×10^{4}
GrowthRateSchmidt2016	E. coli	3	2	6	fitness	scalar	global	7.8×10^{3}
GrowthWang2015	E. coli	46	1	46	fitness	scalar	global	2.1×10^{4}
TranscriptionFactorKnockoutChoe2019	E. coli	2	1	2	fitness	scalar	global	5.3×10^{3}
FitnessContrastBorchert2023	P. putida	5,031	12	64,853	t_statistic (rbtnseq_gene_fitness_two_sample_t_test)	scalar	global	2.2×10^{7}
Flux								
FluxIshii2007	E. coli	25	5	28	net_flux (c13_mfa_net_flux_percent_of_specific_glucose_uptake)	vector (42)	bipartite node	6.0×10^{4}
FluxLi2021	E. coli	3	1	3	net_flux (c13_mfa_net_flux_mmol_per_gdw_per_h)	vector (99)	bipartite node	5.7×10^{4}
Gene interaction								
GeneInteractionBabu2014	E. coli	42,703	1	42,705	gene_interaction	scalar	edge	1.2×10^{7}
GeneInteractionButland2008	E. coli	314,769	1	314,769	gene_interaction	scalar	edge	8.2×10^{7}
Metabolite								
ExometaboliteBrunk2016	E. coli	9	28	126	metabolite_level (hplc_extracellular_concentration_g_per_l)	vector (6)	bipartite node	1.8×10^{5}
MetaboliteIntensityRapp2026	E. coli	407	1	407	metabolite_level (fi_ms_annotated_feature_absolute_intensity_mean_of_2_plates)	vector (2)	bipartite node	3.9×10^{5}
MetabolomeBrunk2016	E. coli	9	26	117	metabolite_level (lc_ms_concentration_um)	vector (51)	bipartite node	3.5×10^{5}
MetabolomeFuhrer2017	E. coli	3,806	1	3,806	metabolite_level (fia_tof_ms_ion_modified_z_score)	vector (7534)	bipartite node	1.9×10^{9}
MetabolomeIshii2007	E. coli	25	5	28	metabolite_level (ce_tofms_intracellular_concentration_mm)	vector (130)	bipartite node	4.9×10^{5}
MetabolomeRapp2026	E. coli	1,497	1	1,497	metabolite_level (fi_ms_iml1515_feature_fold_change_vs_batch_median_mean_of_2_plates)	vector (1321)	bipartite node	5.6×10^{8}
MetabolomeSchastnaya2021	E. coli	103	5	200	metabolite_level (fia_tof_ms_ion_log2_fold_change_vs_wild_type)	vector (284)	bipartite node	3.6×10^{6}
TargetedMetabolomeRapp2026	E. coli	407	1	407	metabolite_level (lc_msms_eic_peak_height_fold_change_vs_library_median)	vector (2)	bipartite node	3.7×10^{5}
MetaboliteGrowthPhaseMenasalvas2025	P. putida	3	1	3	metabolite_level (lc_ms_intracellular_concentration_uM)	vector (46)	bipartite node	7.4×10^{4}
MetaboliteProductionPhaseMenasalvas2025	P. putida	3	1	3	metabolite_level (lc_ms_intracellular_concentration_uM)	vector (46)	bipartite node	7.3×10^{4}
Morphology								
MorphologyCampos2018	E. coli	3,664	1	3,664	morphology	vector (15)	global	8.5×10^{6}
Phosphoproteome								
PhosphositeIntensityPotel2018	E. coli	1	2	12	phosphosite_intensity (fe3_imac_lc_msms_maxquant_summed_xic)	vector (989)	node	1.3×10^{7}
Product titer								
BiofuelTiterBrunk2016	E. coli	8	18	72	titer	scalar	global	1.1×10^{5}
IsopentenolTiterFoo2014	E. coli	9	1	9	titer	scalar	global	1.5×10^{4}
PineneFoldChangeNiu2019	E. coli	40	1	40	titer	scalar	global	2.2×10^{4}
IsoprenolTiterBanerjee2024	P. putida	14	9	20	titer	scalar	global	3.4×10^{4}
IsoprenolTiterCarruthers2025	P. putida	432	2	502	titer	scalar	global	4.2×10^{5}
IsoprenolTiterDeSiqueira2025	P. putida	5	2	10	titer	scalar	global	2.1×10^{5}
IsoprenolTiterYunus2026	P. putida	125	1	125	titer	scalar	global	6.6×10^{4}
IsoprenylAcetateTiterKang2026	P. putida	11	13	26	titer	scalar	global	6.7×10^{4}
ValerolactamTiterThompson2019	P. putida	4	2	8	titer	scalar	global	1.7×10^{4}
Promoter activity								
PromoterReporterMohiuddin2022	E. coli	1,809	27	69,480	promoter_activity	scalar	node	2.8×10^{7}
Protein abundance								
ProteinFoldChangeCaglar2017	E. coli	1	8	8	protein_fold_change (deseq2_wald_log2_fold_change_design_batch_plus_condition)	vector (4180)	node	1.1×10^{7}
ProteomeBrunk2016	E. coli	9	18	81	protein_abundance (srm_protein_peak_area)	vector (68)	node	3.6×10^{5}
ProteomeCaglar2017	E. coli	1	40	105	protein_abundance (lcmsms_spectral_count_deseq2_size_factor_normalized)	vector (4196)	node	2.5×10^{7}
ProteomeIshii2007	E. coli	25	5	28	protein_abundance (lc_ms_ms_absolute_mg_protein_per_g_dry_cell_weight)	vector (57)	node	2.9×10^{5}
ProteomeMori2021	E. coli	1	1	7	protein_abundance (absolute_protein_mass_fraction_xtop_dia_swath_riboprofiling_scaled)	vector (1897)	node	1.1×10^{6}
ProteomeSchmidt2016	E. coli	1	21	21	protein_abundance (absolute_protein_copies_per_cell_label_free_sid_anchored)	vector (2359)	node	1.1×10^{7}
ProteomeSrmSet1Schmidt2016	E. coli	1	18	18	protein_abundance (absolute_protein_copies_per_cell_srm_sid_dataset1_technical_sd)	vector (41)	node	1.5×10^{5}
ProteomeSrmSet2Schmidt2016	E. coli	1	21	21	protein_abundance (absolute_protein_copies_per_cell_srm_sid_dataset2_biological_sd)	vector (31)	node	1.3×10^{5}
CampaignProteomeCarruthers2025	P. putida	401	1	465	protein_abundance (dia_nn_top3_percent_of_proteome_mean)	vector (1616)	node	1.3×10^{8}
CrispriArrayYunus2026	P. putida	25	1	25	protein_fold_change (dia_nn_top3_relative_to_control_strain)	scalar	node	2.5×10^{4}
CrispriDifferentialProteomeYunus2026	P. putida	1	1	1	protein_fold_change (dia_nn_top3_fold_change_relative_to_control_strain_equal_variance_t_test)	vector (305)	node	6.1×10^{4}
CrispriKnockdownYunus2026	P. putida	102	1	102	protein_fold_change (dia_nn_top3_relative_to_control_strain)	scalar	node	4.9×10^{4}
CrispriPanelProteomeYunus2026	P. putida	125	1	125	protein_abundance (dia_nn_top3_signal_benchling_displayed)	vector (207)	node	1.5×10^{6}
ProteomeBanerjee2025	P. putida	4	3	6	protein_abundance (diann_lfq_log2_intensity_replicate_mean)	vector (2248)	node	2.8×10^{6}
ProteomeCarruthers2025	P. putida	17	2	21	protein_abundance (dia_nn_top3_peptide_signal_mean)	vector (1424)	node	5.6×10^{6}
ProteomeDeSiqueira2025	P. putida	6	3	17	protein_abundance (dia_nn_top3_peptide_signal_replicate_mean)	vector (1531)	node	5.5×10^{6}
ProteomeFoldChangeCarruthers2025	P. putida	16	1	16	protein_fold_change (dia_log2_fold_change_paired_two_tailed_t_test)	vector (350)	node	1.5×10^{6}
ProteomeFoldChangeLim2025	P. putida	2	2	4	protein_fold_change (dia_nn_top3_log2_fold_change_welch_two_sample_t_test)	vector (2361)	node	3.7×10^{6}
ProteomeLim2025	P. putida	3	1	3	protein_abundance (dia_nn_top3_log2_mean)	vector (2361)	node	1.5×10^{6}
ProteomeLog10PercentDeSiqueira2025	P. putida	6	3	17	protein_abundance (dia_nn_top3_log10_percent_of_total_abundance_replicate_mean)	vector (1531)	node	6.0×10^{6}
ProteomeMenasalvas2025	P. putida	3	2	6	protein_abundance (dia_protein_counts_sum_mean)	vector (2090)	node	2.6×10^{6}
ProteomePercentDeSiqueira2025	P. putida	6	3	17	protein_abundance (dia_nn_top3_percent_of_total_abundance_replicate_mean)	vector (1531)	node	6.0×10^{6}
Protein synthesis / turnover								
ProteinSynthesisRateLi2014	E. coli	1	3	3	synthesis_rate (ribosome_profiling_footprint_density_absolute_synthesis_rate)	vector (3041)	node	3.7×10^{5}
ProteinTurnoverGupta2024	E. coli	6	8	13	degradation_rate (n15_ammonium_tmtproc_total_turnover_rate_per_hour)	vector (2553)	node	1.1×10^{7}
Viability								
GeneEssentialityBaba2006	E. coli	4,288	1	4,288	is_essential	scalar	node	1.8×10^{6}
GeneEssentialityGerdes2003	E. coli	3,739	1	3,739	is_essential	scalar	node	1.9×10^{6}
GeneEssentialityGoodall2018	E. coli	8,309	2	8,309	is_essential	scalar	node	5.1×10^{6}
GeneEssentialityPrice2018Ecoli	E. coli	324	1	324	is_essential	scalar	node	2.6×10^{5}
GradedGeneEssentialityGoodall2018	E. coli	317	2	317	essentiality_call	scalar	node	2.2×10^{5}
GradedGeneEssentialityRoyet2025	P. putida	5,729	2	11,458	essentiality_call	scalar	node	4.8×10^{6}
Total (100 datasets, 100 probed)				**7,257,468**				7.7×10^{9}

Table 2: Signal controls, same pass. *Structure*: leaves replaced by type tokens. *Shuffled*: numeric leaves permuted within key path. *Noise*: digits of numeric leaves redrawn uniformly at stored precision. *Binary*: per-path z-score thresholded at 0, written in place. *Packed 1-bit*: gzip of those bits alone, 8 per byte.

Dataset	Instances	Numeric leaves	Signal (gzip, bits)	Structure (bits)	Shuffled (bits)	Noise (bits)	Binary (bits)	Packed 1-bit (bits)	Structure/Signal	Signal/Noise	Packed/Signal
Environmental / chemogenomic response											
CrisprPineneToleranceNiu2019	54	108	4.0×10^{4}	7.8×10^{3}	4.0×10^{4}	4.2×10^{4}	3.5×10^{4}	2.7×10^{2}	0.20	0.94	0.01
CrispriChemgenChoe2025	470,502	1,411,506	2.4×10^{8}	2.9×10^{7}	2.5×10^{8}	2.5×10^{8}	1.9×10^{8}	8.3×10^{5}	0.12	0.94	0.00
CrispriGeneFitnessWang2018	20,158	121,191	1.5×10^{7}	2.2×10^{6}	1.5×10^{7}	1.7×10^{7}	8.7×10^{6}	1.1×10^{5}	0.14	0.92	0.01
CrispriGuideFfaEnrichmentFang2025	15,711	47,613	7.3×10^{6}	1.7×10^{6}	7.3×10^{6}	7.8×10^{6}	5.9×10^{6}	2.4×10^{4}	0.23	0.93	0.00
CrispriGuideFitnessWang2018	240,481	963,737	1.6×10^{8}	2.6×10^{7}	1.6×10^{8}	1.7×10^{8}	1.2×10^{8}	4.2×10^{5}	0.16	0.94	0.00
CrispriKnockdownCui2018	143,392	441,964	7.1×10^{7}	1.0×10^{7}	7.1×10^{7}	7.7×10^{7}	5.7×10^{7}	1.9×10^{5}	0.15	0.93	0.00
CrispriScreenRousset2018	92,252	300,128	4.1×10^{7}	5.5×10^{6}	4.2×10^{7}	4.5×10^{7}	3.0×10^{7}	9.3×10^{4}	0.13	0.91	0.00
DoublingTimeCaglar2017	55	275	3.2×10^{4}	5.5×10^{3}	3.2×10^{4}	3.5×10^{4}	2.0×10^{4}	4.5×10^{2}	0.17	0.90	0.01
EnvChemgenGirgis2009	66,287	132,574	3.6×10^{7}	4.8×10^{6}	3.7×10^{7}	3.9×10^{7}	3.4×10^{7}	9.1×10^{4}	0.13	0.94	0.00
EnvChemgenNichols2011	1,286,962	2,573,924	4.0×10^{8}	7.2×10^{7}	4.0×10^{8}	4.2×10^{8}	3.4×10^{8}	1.5×10^{6}	0.18	0.96	0.00
EnvChemgenShiver2016	210,998	210,998	9.9×10^{7}	1.7×10^{7}	9.9×10^{7}	10.0×10^{7}	8.2×10^{7}	2.1×10^{5}	0.18	0.99	0.00
EnvChemgenWang2015	46	92	2.9×10^{4}	7.4×10^{3}	2.9×10^{4}	2.9×10^{4}	2.4×10^{4}	2.6×10^{2}	0.26	0.98	0.01
EnvChemgenWang2024	26,514	106,056	9.3×10^{6}	1.9×10^{6}	9.4×10^{6}	1.1×10^{7}	7.3×10^{6}	6.7×10^{4}	0.20	0.85	0.01
FitnessBrowserEcoli	22,734	68,202	1.1×10^{7}	1.5×10^{6}	1.1×10^{7}	1.1×10^{7}	4.6×10^{6}	6.0×10^{4}	0.13	0.98	0.01
GrowthRateLamoureux2023	354	354	1.9×10^{5}	3.8×10^{4}	1.9×10^{5}	2.0×10^{5}	1.8×10^{5}	5.3×10^{2}	0.20	0.98	0.00
GrowthRateS23Schmidt2016	26	92	2.0×10^{4}	4.6×10^{3}	2.0×10^{4}	2.3×10^{4}	1.8×10^{4}	2.5×10^{2}	0.23	0.89	0.01
GrowthYieldBaba2006	11,735	11,735	3.3×10^{6}	1.4×10^{6}	3.3×10^{6}	3.4×10^{6}	3.0×10^{6}	8.7×10^{3}	0.41	0.96	0.00
MismatchCrispriFitnessHawkins2020	24,186	117,184	1.5×10^{7}	1.5×10^{6}	1.5×10^{7}	1.6×10^{7}	8.1×10^{6}	6.3×10^{4}	0.10	0.92	0.00
PhageRbTnseqMutalik2020	287,815	1,151,260	2.1×10^{8}	2.4×10^{7}	2.3×10^{8}	2.4×10^{8}	1.2×10^{8}	7.1×10^{5}	0.11	0.90	0.00
RbTnseqPrice2018Ecoli	598,662	2,394,648	4.0×10^{8}	5.0×10^{7}	4.5×10^{8}	4.7×10^{8}	2.2×10^{8}	1.5×10^{6}	0.12	0.86	0.00
RbTnseqTeteneva2024	66,162	132,324	2.3×10^{7}	3.7×10^{6}	2.3×10^{7}	2.4×10^{7}	1.7×10^{7}	8.9×10^{4}	0.16	0.96	0.00
RbTnseqWetmore2015	353,662	1,414,648	2.4×10^{8}	2.9×10^{7}	2.6×10^{8}	2.7×10^{8}	1.3×10^{8}	8.4×10^{5}	0.12	0.90	0.00
EnvMetalTnseqRoyet2025	21,583	86,332	6.7×10^{6}	1.5×10^{6}	6.7×10^{6}	8.2×10^{6}	5.5×10^{6}	4.9×10^{4}	0.23	0.82	0.01
FitnessBrowserPutida	1,118,052	3,354,156	5.3×10^{8}	7.2×10^{7}	5.3×10^{8}	5.4×10^{8}	2.1×10^{8}	3.1×10^{6}	0.13	0.98	0.01
IsoprenolSelectionMenasalvas2025	58	466	7.5×10^{4}	2.8×10^{4}	7.5×10^{4}	9.0×10^{4}	7.2×10^{4}	1.9×10^{2}	0.37	0.83	0.00
IsoprenolToleranceLim2025	3	26	1.8×10^{4}	6.4×10^{3}	1.8×10^{4}	1.9×10^{4}	1.8×10^{4}	1.9×10^{2}	0.35	0.96	0.01
LactamGrowthRateThompson2019	9	9	1.2×10^{4}	4.6×10^{3}	1.2×10^{4}	1.2×10^{4}	1.1×10^{4}	1.8×10^{2}	0.39	0.99	0.02
RbTnseqBorchert2023	10,824	10,824	4.5×10^{6}	1.1×10^{6}	4.5×10^{6}	4.5×10^{6}	3.4×10^{6}	9.2×10^{3}	0.25	0.99	0.00
RbTnseqBorchert2024	1,466,920	2,420,011	4.2×10^{8}	1.1×10^{8}	4.2×10^{8}	4.3×10^{8}	2.8×10^{8}	2.4×10^{6}	0.26	0.98	0.01
Expression (RNA-seq)											
MrnaFractionBalakrishnan2022	28	121,604	7.2×10^{6}	2.4×10^{6}	7.2×10^{6}	8.3×10^{6}	2.7×10^{6}	1.1×10^{5}	0.33	0.86	0.01
RnaseqCaglar2017	152	1,275,584	6.7×10^{7}	2.5×10^{7}	7.1×10^{7}	7.2×10^{7}	2.7×10^{7}	9.5×10^{5}	0.37	0.93	0.01
RnaseqGuzman2018	14	60,678	4.5×10^{6}	1.3×10^{6}	4.5×10^{6}	4.5×10^{6}	1.5×10^{6}	6.0×10^{4}	0.29	0.98	0.01
RnaseqLamoureux2023	1,035	8,811,990	5.8×10^{8}	1.7×10^{8}	5.9×10^{8}	5.8×10^{8}	1.9×10^{8}	7.3×10^{6}	0.29	1.00	0.01
RnaseqPublicK12Lamoureux2023	1,675	14,590,925	9.5×10^{8}	2.8×10^{8}	9.7×10^{8}	9.5×10^{8}	3.0×10^{8}	1.1×10^{7}	0.30	0.99	0.01
PutidaPrecise321Lim2022	321	3,572,400	2.4×10^{8}	7.0×10^{7}	2.4×10^{8}	2.4×10^{8}	7.7×10^{7}	2.2×10^{6}	0.29	1.00	0.01
Fitness											
CarbonSourceTong2020	111,420	220,740	4.8×10^{7}	3.6×10^{6}	4.8×10^{7}	5.0×10^{7}	4.0×10^{7}	1.1×10^{5}	0.07	0.96	0.00
CrispriCrossRachwalski2024	48,900	136,563	1.9×10^{7}	4.0×10^{6}	1.9×10^{7}	2.1×10^{7}	1.4×10^{7}	6.6×10^{4}	0.21	0.91	0.00
GrowthAucRapp2026	1,515	7,575	7.7×10^{5}	4.9×10^{4}	7.7×10^{5}	8.3×10^{5}	3.5×10^{5}	5.1×10^{3}	0.06	0.92	0.01
GrowthRateCampos2018	3,664	7,328	1.1×10^{6}	1.5×10^{5}	1.1×10^{6}	1.1×10^{6}	7.4×10^{5}	4.2×10^{3}	0.14	0.97	0.00
GrowthRateChoe2019	8	32	1.1×10^{4}	2.9×10^{3}	1.1×10^{4}	1.1×10^{4}	8.1×10^{3}	1.9×10^{2}	0.28	0.99	0.02
GrowthRateSchmidt2016	6	24	7.8×10^{3}	2.5×10^{3}	7.8×10^{3}	7.9×10^{3}	6.0×10^{3}	1.8×10^{2}	0.32	0.99	0.02
GrowthWang2015	46	184	2.1×10^{4}	3.7×10^{3}	2.0×10^{4}	2.7×10^{4}	1.2×10^{4}	3.5×10^{2}	0.18	0.76	0.02
TranscriptionFactorKnockoutChoe2019	2	2	5.3×10^{3}	2.4×10^{3}	5.3×10^{3}	5.3×10^{3}	5.2×10^{3}	1.7×10^{2}	0.45	1.00	0.03
FitnessContrastBorchert2023	64,853	389,118	2.2×10^{7}	2.1×10^{6}	2.8×10^{7}	3.3×10^{7}	9.2×10^{6}	1.4×10^{5}	0.10	0.66	0.01
Flux											
FluxIshii2007	28	1,200	6.0×10^{4}	10.0×10^{3}	6.2×10^{4}	7.0×10^{4}	2.7×10^{4}	1.4×10^{3}	0.17	0.86	0.02
FluxLi2021	3	1,104	5.7×10^{4}	8.5×10^{3}	6.1×10^{4}	9.8×10^{4}	1.8×10^{4}	1.3×10^{3}	0.15	0.58	0.02
Gene interaction											
GeneInteractionBabu2014	42,705	85,410	1.2×10^{7}	2.8×10^{6}	1.2×10^{7}	1.2×10^{7}	9.7×10^{6}	4.4×10^{4}	0.25	0.93	0.00
GeneInteractionButland2008	314,769	629,538	8.2×10^{7}	2.3×10^{7}	8.2×10^{7}	8.7×10^{7}	6.6×10^{7}	4.6×10^{5}	0.28	0.94	0.01
Metabolite											
ExometaboliteBrunk2016	126	2,450	1.8×10^{5}	3.7×10^{4}	1.8×10^{5}	2.2×10^{5}	1.4×10^{5}	1.9×10^{3}	0.21	0.78	0.01
MetaboliteIntensityRapp2026	407	4,532	3.9×10^{5}	1.3×10^{5}	3.9×10^{5}	4.5×10^{5}	2.7×10^{5}	4.1×10^{3}	0.32	0.88	0.01
MetabolomeBrunk2016	117	8,511	3.5×10^{5}	5.2×10^{4}	3.5×10^{5}	4.8×10^{5}	1.8×10^{5}	5.0×10^{3}	0.15	0.71	0.01
MetabolomeFuhrer2017	3,806	57,348,808	1.9×10^{9}	1.1×10^{9}	1.9×10^{9}	2.1×10^{9}	1.1×10^{9}	2.9×10^{7}	0.56	0.90	0.02
MetabolomeIshii2007	28	8,080	4.9×10^{5}	5.1×10^{4}	4.9×10^{5}	5.8×10^{5}	1.1×10^{5}	4.7×10^{3}	0.10	0.86	0.01
MetabolomeRapp2026	1,497	5,934,108	5.6×10^{8}	1.1×10^{8}	5.6×10^{8}	5.8×10^{8}	1.9×10^{8}	3.7×10^{6}	0.19	0.96	0.01
MetabolomeSchastnaya2021	200	116,436	3.6×10^{6}	4.7×10^{5}	3.7×10^{6}	5.3×10^{6}	1.6×10^{6}	6.2×10^{4}	0.13	0.68	0.02
TargetedMetabolomeRapp2026	407	2,899	3.7×10^{5}	1.0×10^{5}	3.7×10^{5}	3.8×10^{5}	2.4×10^{5}	2.4×10^{3}	0.28	0.96	0.01
MetaboliteGrowthPhaseMenasalvas2025	3	448	7.4×10^{4}	1.4×10^{4}	7.5×10^{4}	8.0×10^{4}	6.4×10^{4}	4.8×10^{2}	0.20	0.92	0.01
MetaboliteProductionPhaseMenasalvas2025	3	450	7.3×10^{4}	1.5×10^{4}	7.4×10^{4}	8.0×10^{4}	6.4×10^{4}	4.9×10^{2}	0.20	0.92	0.01
Morphology											
MorphologyCampos2018	3,664	97,234	8.5×10^{6}	1.8×10^{5}	8.5×10^{6}	9.5×10^{6}	1.1×10^{6}	9.2×10^{4}	0.02	0.90	0.01
Phosphoproteome											
PhosphositeIntensityPotel2018	12	141,757	1.3×10^{7}	2.7×10^{6}	1.3×10^{7}	1.4×10^{7}	7.3×10^{6}	1.5×10^{4}	0.21	0.94	0.00
Product titer											
BiofuelTiterBrunk2016	72	702	1.1×10^{5}	3.0×10^{4}	1.1×10^{5}	1.3×10^{5}	1.0×10^{5}	4.0×10^{2}	0.28	0.84	0.00
IsopentenolTiterFoo2014	9	36	1.5×10^{4}	5.4×10^{3}	1.5×10^{4}	1.6×10^{4}	1.5×10^{4}	2.0×10^{2}	0.35	0.94	0.01
PineneFoldChangeNiu2019	40	160	2.2×10^{4}	6.1×10^{3}	2.2×10^{4}	2.9×10^{4}	2.1×10^{4}	3.2×10^{2}	0.28	0.77	0.01
IsoprenolTiterBanerjee2024	20	159	3.4×10^{4}	1.1×10^{4}	3.4×10^{4}	3.8×10^{4}	3.3×10^{4}	3.0×10^{2}	0.32	0.89	0.01
IsoprenolTiterCarruthers2025	502	6,105	4.2×10^{5}	1.2×10^{5}	4.2×10^{5}	5.4×10^{5}	2.7×10^{5}	2.9×10^{3}	0.30	0.77	0.01
IsoprenolTiterDeSiqueira2025	10	1,182	2.1×10^{5}	3.1×10^{4}	2.1×10^{5}	2.3×10^{5}	1.8×10^{5}	1.4×10^{3}	0.15	0.90	0.01
IsoprenolTiterYunus2026	125	500	6.6×10^{4}	1.4×10^{4}	6.7×10^{4}	7.6×10^{4}	5.9×10^{4}	2.9×10^{2}	0.22	0.87	0.00
IsoprenylAcetateTiterKang2026	26	329	6.7×10^{4}	1.9×10^{4}	6.7×10^{4}	7.5×10^{4}	6.5×10^{4}	3.1×10^{2}	0.29	0.89	0.00
ValerolactamTiterThompson2019	8	40	1.7×10^{4}	6.6×10^{3}	1.7×10^{4}	1.8×10^{4}	1.7×10^{4}	2.0×10^{2}	0.38	0.96	0.01
Promoter activity											
PromoterReporterMohiuddin2022	69,480	208,440	2.8×10^{7}	3.6×10^{6}	2.8×10^{7}	3.2×10^{7}	2.6×10^{7}	4.1×10^{4}	0.13	0.88	0.00
Protein abundance											
ProteinFoldChangeCaglar2017	8	154,960	1.1×10^{7}	3.0×10^{6}	1.2×10^{7}	1.2×10^{7}	3.2×10^{6}	1.1×10^{5}	0.27	0.91	0.01
ProteomeBrunk2016	81	11,619	3.6×10^{5}	3.4×10^{4}	3.6×10^{5}	5.3×10^{5}	1.3×10^{5}	6.8×10^{3}	0.09	0.68	0.02
ProteomeCaglar2017	105	881,160	2.5×10^{7}	1.7×10^{7}	3.7×10^{7}	4.2×10^{7}	1.8×10^{7}	3.8×10^{5}	0.67	0.60	0.01
ProteomeIshii2007	28	4,722	2.9×10^{5}	1.2×10^{4}	3.0×10^{5}	3.5×10^{5}	3.9×10^{4}	3.9×10^{3}	0.04	0.81	0.01
ProteomeMori2021	7	26,716	1.1×10^{6}	1.1×10^{5}	1.1×10^{6}	1.3×10^{6}	4.0×10^{5}	1.4×10^{4}	0.10	0.84	0.01
ProteomeSchmidt2016	21	145,005	1.1×10^{7}	2.7×10^{5}	1.1×10^{7}	1.1×10^{7}	1.5×10^{6}	9.0×10^{4}	0.02	1.00	0.01
ProteomeSrmSet1Schmidt2016	18	2,214	1.5×10^{5}	4.7×10^{3}	1.5×10^{5}	1.6×10^{5}	2.1×10^{4}	2.0×10^{3}	0.03	0.92	0.01
ProteomeSrmSet2Schmidt2016	21	1,953	1.3×10^{5}	4.1×10^{3}	1.3×10^{5}	1.4×10^{5}	2.0×10^{4}	1.9×10^{3}	0.03	0.93	0.01
CampaignProteomeCarruthers2025	465	1,978,591	1.3×10^{8}	7.8×10^{6}	1.3×10^{8}	1.5×10^{8}	2.9×10^{7}	1.4×10^{6}	0.06	0.88	0.01
CrispriArrayYunus2026	25	225	2.5×10^{4}	7.5×10^{3}	2.5×10^{4}	2.8×10^{4}	1.8×10^{4}	3.9×10^{2}	0.30	0.88	0.02
CrispriDifferentialProteomeYunus2026	1	916	6.1×10^{4}	1.1×10^{4}	6.1×10^{4}	6.6×10^{4}	1.8×10^{4}	1.9×10^{2}	0.19	0.92	0.00
CrispriKnockdownYunus2026	102	306	4.9×10^{4}	1.5×10^{4}	4.9×10^{4}	5.2×10^{4}	4.5×10^{4}	2.4×10^{2}	0.31	0.93	0.00
CrispriPanelProteomeYunus2026	125	51,875	1.5×10^{6}	7.0×10^{4}	1.5×10^{6}	2.1×10^{6}	2.9×10^{5}	2.8×10^{4}	0.05	0.69	0.02
ProteomeBanerjee2025	6	40,239	2.8×10^{6}	1.5×10^{5}	2.8×10^{6}	3.0×10^{6}	6.8×10^{5}	2.8×10^{4}	0.05	0.91	0.01
ProteomeCarruthers2025	21	81,310	5.6×10^{6}	2.6×10^{5}	5.7×10^{6}	6.3×10^{6}	1.3×10^{6}	5.2×10^{4}	0.05	0.90	0.01
ProteomeDeSiqueira2025	17	79,637	5.5×10^{6}	6.6×10^{5}	5.6×10^{6}	6.2×10^{6}	1.5×10^{6}	5.4×10^{4}	0.12	0.89	0.01
ProteomeFoldChangeCarruthers2025	16	17,970	1.5×10^{6}	1.3×10^{5}	1.5×10^{6}	1.5×10^{6}	2.6×10^{5}	1.3×10^{4}	0.09	0.98	0.01
ProteomeFoldChangeLim2025	4	47,251	3.7×10^{6}	10.0×10^{5}	3.8×10^{6}	4.0×10^{6}	1.1×10^{6}	3.8×10^{4}	0.27	0.93	0.01
ProteomeLim2025	3	21,240	1.5×10^{6}	4.5×10^{5}	1.5×10^{6}	1.6×10^{6}	4.9×10^{5}	1.5×10^{4}	0.30	0.93	0.01
ProteomeLog10PercentDeSiqueira2025	17	79,637	6.0×10^{6}	6.6×10^{5}	6.0×10^{6}	6.2×10^{6}	1.6×10^{6}	5.4×10^{4}	0.11	0.96	0.01
ProteomeMenasalvas2025	6	37,964	2.6×10^{6}	3.6×10^{5}	2.7×10^{6}	3.0×10^{6}	7.5×10^{5}	2.6×10^{4}	0.14	0.89	0.01
ProteomePercentDeSiqueira2025	17	79,637	6.0×10^{6}	6.6×10^{5}	6.0×10^{6}	6.2×10^{6}	1.5×10^{6}	5.4×10^{4}	0.11	0.96	0.01
Protein synthesis / turnover											
ProteinSynthesisRateLi2014	3	8,647	3.7×10^{5}	1.7×10^{5}	3.7×10^{5}	4.2×10^{5}	2.0×10^{5}	7.7×10^{3}	0.45	0.89	0.02
ProteinTurnoverGupta2024	13	141,301	1.1×10^{7}	2.9×10^{6}	1.1×10^{7}	1.1×10^{7}	3.6×10^{6}	9.6×10^{4}	0.27	0.99	0.01
Viability											
GeneEssentialityBaba2006	4,288	17,152	1.8×10^{6}	5.0×10^{5}	1.8×10^{6}	2.1×10^{6}	1.8×10^{6}	9.0×10^{3}	0.28	0.86	0.00
GeneEssentialityGerdes2003	3,739	13,858	1.9×10^{6}	5.4×10^{5}	1.9×10^{6}	1.9×10^{6}	1.6×10^{6}	1.2×10^{4}	0.28	1.00	0.01
GeneEssentialityGoodall2018	8,309	24,927	5.1×10^{6}	1.1×10^{6}	5.1×10^{6}	5.5×10^{6}	3.9×10^{6}	1.8×10^{4}	0.22	0.93	0.00
GeneEssentialityPrice2018Ecoli	324	2,916	2.6×10^{5}	5.2×10^{4}	2.6×10^{5}	3.3×10^{5}	2.2×10^{5}	2.6×10^{3}	0.20	0.80	0.01
GradedGeneEssentialityGoodall2018	317	951	2.2×10^{5}	4.6×10^{4}	2.2×10^{5}	2.3×10^{5}	1.7×10^{5}	1.1×10^{3}	0.22	0.93	0.01
GradedGeneEssentialityRoyet2025	11,458	103,122	4.8×10^{6}	9.4×10^{5}	4.9×10^{6}	6.5×10^{6}	3.6×10^{6}	6.7×10^{4}	0.19	0.74	0.01
Total (100 probed)	**7,257,468**	**115,325,823**	7.7×10^{9}	2.3×10^{9}	7.9×10^{9}	8.2×10^{9}	4.1×10^{9}	7.0×10^{7}	**0.29**	**0.94**	**0.01**